Integral Transform

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1 Integral Transform

I Integral Transform

i Integral Transform

An integral transform is in the form

$$\int_a^b K(s,t)f(t)\,\mathrm{d}t$$

where $K(s,t)$ is called kernel.

ii Linear transform

An integral transform

$$\int_a^b K(s,t)f(t)\,\mathrm{d}t$$

is a linear transform iff

$$C\int_a^b K(s,t)f(t)\,\mathrm{d}t+D\int_a^b K(s,t)g(t)\,\mathrm{d}t+E\int_a^b K(s,t)h(t)\,\mathrm{d}t=\int_a^b K(s,t)(Cf(t)+Dg(t)+Eh(t))\,\mathrm{d}t$$

whenever the LHS exists.

II Laplace Transform (拉普拉斯變換)

i Introduction

The one-sided or unilateral Laplace transform and two-sided or bilateral Laplace transform, where Laplace transform may refer to either of them or both of them depending on context and often refer to the former when not clear, are two integral transform that convert a complex-valued signal, for both one-sided and two-sided, often denoted in lowercase such as f or $f(t)$, of a real variable t, called time domain, to a complex-valued function, often denoted in uppercase of the same letter, F or $F(s)$, or as $\mathcal{L}\{f\}$, $\mathcal{L}\{f(t)\}$, $\mathcal{L}\{f\}(s)$, or $\mathcal{L}\{f(t)\}(s)$, and for two-sided, also denoted as $\mathcal{B}\{f\}$, $\mathcal{B}\{f(t)\}$, $\mathcal{B}\{f\}(s)$, or $\mathcal{B}\{f(t)\}(s)$, of a complex variable s, called (complex) frequency domain, (complex) frequency plane, s-domain, s-plane, Laplace domain, or Laplace plane.

ii Definition

Let R be $\mathbb{R}_{\geq 0}$ for one-sided Laplace transform and $\mathbb{R}$ for two-sided Laplace transform.

For any complex-valued (or real-valued) functional f with support on R, the Laplace transform is defined as

$$\int_R e^{-st}f(t)\,\mathrm{d}t\,,$$

of which convergence is defined as the finiteness of the integral, on the subset of $\mathbb{C}$ where it converges.

For any complex-valued (or real-valued) distribution $T \in \mathcal{D}(\mathbb{R})$ with compact support on R, the Laplace transform is defined as

$$\langle T, e^{-st} \rangle,$$

which is always convergent, on $\mathbb{C}$.

Distributions are often denoted in function inside integral notation, notably such as Dirac distribution, but should be interpreted in distributional definition.

Laplace transform and derivative of a sum of functional and distribution in function inside integral notation should be interpreted as the sum of the Laplace transform and derivative of it. Function value at a point of a functional and distribution in function inside integral notation should be interpreted as function value of the functional.

For one-sided Laplace transform:

- Restriction of functional or distribution on R is represented with the same symbol of the functional or distribution and interpreted as either depending on context.
- $\int_0^\infty$ is also denoted as $\int_{0^-}^\infty$, i.e., $\lim_{\varepsilon \to 0^+} \int_{-\varepsilon}^\infty$, and 0 as input of a time domain signal is also denoted as 0^-. This notation is intuitive in Riemann–Stieltjes integral interpretation of Dirac delta.

iii Region of convergence (ROC)

The set of values of s for which $F(s)$ converges is called region of convergence (ROC). A function is called Laplace-transformable iff the ROC of its Laplace transform is nonempty.

For one-sided Laplace transform, ROC is a class of the form of either $\Re(s) > a$ or $\Re(s) \geq a$ for some extended real constant a, called abscissa of convergence.

For two-sided Laplace transform, ROC is a class of the form of either $\Re(s) > a$ or $\Re(s) \geq a$ and either $\Re(s) <$ or $\Re(s) \leq b$ for some extended real constants a and b, called abscissas of convergence.

iv Of exponential order

A complex-valued (or real-valued) functional f defined on $\mathbb{R}_{\geq a}$ for some $a \in \mathbb{R}$ is of exponential order $\alpha \in \mathbb{R}_{>0}$ iff there exists $M > 0$ and $T > 0$ such that:

$$|f(t)| \leq M e^{\alpha t} \quad \forall t > T.$$

A complex-valued (or real-valued) functional f defined on $\mathbb{R}_{\geq a}$ for some $a \in \mathbb{R}$ is of strict exponential order $\alpha \in \mathbb{R}_{>0}$ iff there exists $M > 0$ and $T > 0$ such that:

$$|f(t)| < Me^{\alpha t} \quad \forall t > T.$$

A complex-valued (or real-valued) functional f defined on $\mathbb{R}_{\leq a}$ for some $a \in \mathbb{R}$ is of exponential order $\alpha \in \mathbb{R}_{<0}$ iff there exists $M > 0$ and $T < 0$ such that:

$$|f(t)| \leq Me^{\alpha t} \quad \forall t < T.$$

A complex-valued (or real-valued) functional f defined on $\mathbb{R}_{\leq a}$ for some $a \in \mathbb{R}$ is of strict exponential order $\alpha \in \mathbb{R}_{<0}$ iff there exists $M > 0$ and $T < 0$ such that:

$$|f(t)| < Me^{\alpha t} \quad \forall t < T.$$

A complex-valued (or real-valued) functional f defined on $\mathbb{R}$ is of exponential order iff both $f_{\mathbb{R}_{\geq 0}}$ and $f_{\mathbb{R}_{\leq 0}}$ are of exponential order.

v Laplace transformability theorem

A complex-valued (or real-valued) functional f defined on R is one-sided Laplace-transformable if it is piecewise continuous on all closed subintervals of R and of exponential order.

Proof. We will show the proof for one-sided Laplace transform, the proof for two-sided Laplace transforme is similar.

Assume that $f(t)$ is piecewise continuous on $[0, \infty)$ and there exists $M > 0$ and $T > 0$ such that:

$$|f(t)| \leq Me^{\alpha t} \quad \forall t > T.$$

For $\Re(s) > \alpha$, since $f(t)$ is piecewise continuous on $[0, \infty)$, there exist a locally finite indexed family $A = \{[a_i, b_i] \mid i \in I\}$ such that for each $i \in I \setminus \{j\}$, the integral

$$\int_{a_i}^{b_i} |f(t)|e^{-st}\,\mathrm{d}t$$

exists and is finite.

Split the integral:

$$\int_0^\infty f(t)e^{-st}\,\mathrm{d}t = \int_0^T f(t)e^{-st}\,\mathrm{d}t + \int_T^\infty f(t)e^{-st}\,\mathrm{d}t\,.$$

For the first integral, by local finiteness, there exists a finite cover of $[0, T]$ that is a subset of A. The sum over finitely many finite integral is finite.

For the second integral, since $|f(t)| \leq Me^{\alpha t}$, we have

$$\int_T^\infty |f(t)|e^{-st}\,\mathrm{d}t \leq \int_T^\infty Me^{\alpha t}e^{-st}\,\mathrm{d}t = M\int_T^\infty e^{-(s-\alpha)t}\,\mathrm{d}t\,,$$

which converges since $\Re(s) > \alpha$. □

For one-sided Laplace transform, the ROC is

- $\Re(s) \geq \alpha$ if f is of strict exponential order α and not of exponential order β for all $0 < \beta < \alpha$, or
- $\Re(s) > \alpha$ if f is of exponential order α, not of strict exponential order α, and not of exponential order β for all $0 < \beta < \alpha$.

For two-sided Laplace transform, the ROC is the intersection of the following two regions:

- First region:
  - $\Re(s) \geq \alpha_1$ if $f_{\mathbb{R}_{\geq 0}}$ is of strict exponential order α_1 and not of exponential order β for all $0 < \beta < \alpha_1$, or
  - $\Re(s) > \alpha_1$ if $f_{\mathbb{R}_{\geq 0}}$ is of exponential order α_1, not of strict exponential order α_1, and not of exponential order β for all $0 < \beta < \alpha_1$.
- Second region:
  - $\Re(s) \leq \alpha_2$ if $f_{\mathbb{R}_{\leq 0}}$ is of strict exponential order α_2 and not of exponential order β for all $0 < \beta < \alpha_2$, or
  - $\Re(s) < \alpha_2$ if $f_{\mathbb{R}_{\leq 0}}$ is of exponential order α_2, not of strict exponential order α_2, and not of exponential order β for all $0 < \beta < \alpha_2$.

vi Linearity

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}, \quad a, b \in \mathbb{C}_{\neq 0}.$$

The ROC is the intersection of the ROC of $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{g(t)\}$.

vii Unit step function

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}, \quad a \in R.$$

The ROC is $\Re(s) > 0$.

viii Delta distribution

$$\mathcal{L}\{\frac{\mathrm{d}^n}{\mathrm{d}t^n}\delta(t-a)\} = s^n e^{-sa}, \quad n \in \mathbb{N}_0 \wedge a \in R.$$

The ROC is $\mathbb{C}$.

ix Exponential function

$$\mathcal{L}\left\{e^{at}\right\} = \frac{1}{s-a}, \quad a \in \mathbb{C}_{\neq 0}.$$

For one-sided Laplace transformer, the ROC is $\Re(s) > \Re(a)$.

For two-sided Laplace transformer, the ROC is $\varnothing$.

x First shifting theorem or frequency shift

$$\mathcal{L}\left\{e^{at}f(t)\right\} = F(s-a), \quad a \in \mathbb{C}_{\neq 0}.$$

For one-sided Laplace transformer, the ROC shifts right by $\Re(a)$.

For two-sided Laplace transformer, the ROC change depends on f.

xi Second shifting theorem, time delay, or time shift

For one-sided Laplace transform:

$$\mathcal{L}\left\{f(t-a)\right\} = e^{-as}F(s), \quad a \in \mathbb{R}_{\geq 0}.$$

The ROC doesn't change.

Proof. For functional:

$$\begin{aligned}\mathcal{L}\left\{f(t-a)\right\} &= \int_a^\infty e^{-st}f(t-a)\,\mathrm{d}t \\ &= \int_0^\infty e^{-s(u+a)}f(u)\,\mathrm{d}u \\ &= e^{-as}F(s)\end{aligned}$$

For distribution:

$$\langle \tau_a T, e^{-st}\rangle = \langle T, e^{-s(t+a)}\rangle = e^{-as}\langle T, e^{-st}.$$

□

For two-sided Laplace transform:

$$\mathcal{L}\left\{f(t-a)\right\} = e^{-as}F(s), \quad a \in \mathbb{R}_{\neq 0}.$$

The ROC doesn't change.

Proof. For functional:

$$\begin{aligned}\mathcal{L}\left\{f(t-a)\right\} &= \int_{-\infty}^\infty e^{-st}f(t-a)\,\mathrm{d}t \\ &= \int_{-\infty}^\infty e^{-s(u+a)}f(u)\,\mathrm{d}u \\ &= e^{-as}F(s)\end{aligned}$$

For distribution:

$$\langle \tau_a T, e^{-st}\rangle = \langle T, e^{-s(t+a)}\rangle = e^{-as}\langle T, e^{-st}.$$

□

xii Differentiation in time domain

For one-sided Laplace transform of functional:

$$\mathcal{L}\left\{f'(t)\right\} = sF(s) + \lim_{t\to\infty} e^{-st} f(t) - f(0),$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f}{\mathrm{d}t^n}(t)\right\} = s^n F(s) + \sum_{i=0}^{n-1} s^{n-1-i}\left(\lim_{t\to\infty} e^{-st}\frac{\mathrm{d}^i f}{\mathrm{d}t^i}(t) - \frac{\mathrm{d}^i f}{\mathrm{d}t^i}(0)\right).$$

Proof.

$$u = e^{-st}, \quad \mathrm{d}u = -se^{-st}\,\mathrm{d}t\,.$$

$$\mathrm{d}v = f'(t)\,\mathrm{d}t\,, \quad v = f(t).$$

$$\begin{aligned}\mathcal{L}\left\{f'(t)\right\} &= \int_0^\infty e^{-st} f'(t)\,\mathrm{d}t \\ &= \left(e^{-st}f(t)\right)\Big|_0^\infty + \int_0^\infty f(t)se^{-st}\,\mathrm{d}t \\ &= sF(s) + \lim_{t\to\infty} e^{-st} f(t) - f(0)\end{aligned}$$

□

If f is of exponential order, $\lim_{t\to\infty} e^{-st} f(t) = 0$,

$$\mathcal{L}\left\{f'(t)\right\} = sF(s) - f(0),$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f}{\mathrm{d}t^n}(t)\right\} = s^n F(s) - \sum_{i=0}^{n-1} s^{n-1-i}\frac{\mathrm{d}^i f}{\mathrm{d}t^i}(0),$$

and the ROC is a superset of that of $F(s)$.

For two-sided Laplace transform of functional:

$$\mathcal{L}\left\{f'(t)\right\} = sF(s) + \lim_{t\to\infty} e^{-st} f(t) - \lim_{t\to-\infty} e^{-st} f(t),$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f}{\mathrm{d}t^n}(t)\right\} = s^n F(s) + \sum_{i=0}^{n-1} s^{n-1-i}\left(\lim_{t\to\infty} e^{-st}\frac{\mathrm{d}^i f}{\mathrm{d}t^i}(t) - \lim_{t\to-\infty} e^{-st}\frac{\mathrm{d}^i f}{\mathrm{d}t^i}(t)\right).$$

Proof.

$$u = e^{-st}, \quad \mathrm{d}u = -se^{-st}\,\mathrm{d}t\,.$$

$$\mathrm{d}v = f'(t)\,\mathrm{d}t\,, \quad v = f(t).$$

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= \int_{-\infty}^\infty e^{-st} f'(t)\,\mathrm{d}t \\ &= \left(e^{-st}f(t)\right)\Big|_{-\infty}^\infty + \int_{-\infty}^\infty f(t)se^{-st}\,\mathrm{d}t \\ &= sF(s) + \lim_{t\to\infty} e^{-st} f(t) - \lim_{t\to-\infty} e^{-st} f(t)\end{aligned}$$

□

If f is of exponential order, $\lim_{t\to-\infty} e^{-st}f(t) = \lim_{t\to\infty} e^{-st}f(t) = 0$,

$$\mathcal{L}\{f'(t)\} = sF(s),$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f}{\mathrm{d}t^n}(t)\right\} = s^n F(s),$$

and the ROC is a superset of that of $F(s)$.

For Laplace transform of distribution:

$$\mathcal{L}\{T'\} = s\mathcal{L}\{T\},$$

$$\mathcal{L}\{\frac{\mathrm{d}^n T}{\mathrm{d}t^n}\} = s^n\mathcal{L}\{T\}.$$

Proof.

$$\begin{aligned}\langle T', e^{-st}\rangle &= -\langle T, -se^{-st}\rangle \\ &= s\langle T, e^{-st}\rangle \\ &= s\mathcal{L}\{T\}\end{aligned}$$

□

xiii Integration in time domain

For one-sided Laplace transform of functional:

$$\mathcal{L}\left\{\int_0^t f(\tau)\,\mathrm{d}\tau\right\} = \frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to\infty} e^{-st}\int_0^t f(\tau)\,\mathrm{d}\tau\,.$$

Proof.

$$\begin{aligned}\mathcal{L}\left\{\int_0^t f(\tau)\,\mathrm{d}\tau\right\} &= \int_0^\infty e^{-st}\int_0^t f(\tau)\,\mathrm{d}\tau\,\mathrm{d}t \\ &= \left(-\frac{1}{s}e^{-st}\int_0^t f(\tau)\,\mathrm{d}\tau\right)\Big|_0^\infty + \int_0^\infty \frac{1}{s}e^{-st}f(t)\,\mathrm{d}t \\ &= \frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to\infty} e^{-st}\int_0^t f(\tau)\,\mathrm{d}\tau\end{aligned}$$

□

If f is of exponential order, $\lim_{t\to\infty} e^{-st}\int_0^t f(\tau)\,\mathrm{d}\tau = 0$ for all s in its ROC,

$$\mathcal{L}\left\{\int_0^t f(\tau)\,\mathrm{d}\tau\right\} = \frac{1}{s}F(s),$$

and the ROC is $F(s)$ if there is a zero at 0 of $F(s)$ and $F(s)\cap\{s\in\mathbb{C}\mid\Re(s)>0\}$ otherwise.

For two-sided Laplace transform of functional:

$$\mathcal{L}\left\{\int_{-\infty}^t f(\tau)\,\mathrm{d}\tau\right\} = \frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to\infty} e^{-st}\int_{-\infty}^t f(\tau)\,\mathrm{d}\tau\,.$$

Proof.

$$\begin{aligned}\mathcal{L}\left\{\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right\} &= \int_{-\infty}^{\infty} e^{-st}\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\,\mathrm{d}t \\ &= \left(-\frac{1}{s}e^{-st}\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right)\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\frac{1}{s}e^{-st}f(t)\,\mathrm{d}t \\ &= \frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to\infty} e^{-st}\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\end{aligned}$$

□

If f is of exponential order, $\lim_{t\to\infty} e^{-st}\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = 0$ for all s in its ROC,

$$\mathcal{L}\left\{\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right\} = \frac{1}{s}F(s),$$

and the ROC is $F(s)$ if there is a zero at 0 of $F(s)$ and $F(s)\cap\{s\in\mathbb{C}\mid\Re(s)>0\}$ otherwise.

$$\mathcal{L}\left\{\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\right\} = -\frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to-\infty} e^{-st}\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\,.$$

Proof.

$$\begin{aligned}\mathcal{L}\left\{\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\right\} &= \int_{-\infty}^{\infty} e^{-st}\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\,\mathrm{d}t \\ &= \left(-\frac{1}{s}e^{-st}\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\right)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\frac{1}{s}e^{-st}f(t)\,\mathrm{d}t \\ &= -\frac{1}{s}F(s) - \frac{1}{s}\lim_{t\to-\infty} e^{-st}\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\end{aligned}$$

□

If f is of exponential order, $\lim_{t\to-\infty} e^{-st}\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau = 0$ for all s in its ROC,

$$\mathcal{L}\left\{\int_{t}^{\infty} f(\tau)\,\mathrm{d}\tau\right\} = -\frac{1}{s}F(s),$$

and the ROC is $F(s)$ if there is a zero at 0 of $F(s)$ and $F(s)\cap\{s\in\mathbb{C}\mid\Re(s)<0\}$ otherwise.

xiv Differentiation in frequency domain

$$\begin{aligned}\mathcal{L}\{tf(t)\} &= -\frac{\mathrm{d}}{\mathrm{d}s}F(s), \\ \mathcal{L}\left\{t^n f(t)\right\} &= (-1)^n\frac{\mathrm{d}^n}{\mathrm{d}s^n}F(s).\end{aligned}$$

Proof. For functional, by Leibniz integral rule,

$$\begin{aligned}-\frac{\mathrm{d}}{\mathrm{d}s}F(s) &= -\frac{\mathrm{d}}{\mathrm{d}s}\int_R e^{-st}f(t)\,\mathrm{d}t \\ &= -\int_R \frac{\partial}{\partial s}\left(e^{-st}f(t)\right)\mathrm{d}t \\ &= -\int_R -te^{-st}f(t)\,\mathrm{d}t \\ &= \mathcal{L}\{tf(t)\}\end{aligned}$$

For distribution:

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{L}\{T\} = \langle T, -te^{-st}\rangle = -\mathcal{L}\{tT\}.$$

□

xv Scaling in time domain

For one-sided Laplace transform:

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right), \quad a > 0.$$

For two-sided Laplace transform:

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right), \quad a \neq 0.$$

Proof. For functional:

$$u = at, \quad \mathrm{d}t = \frac{1}{a}\,\mathrm{d}u$$

$$\int_R e^{-st} f(at)\,\mathrm{d}t = \frac{1}{a}\int_R e^{-\frac{s}{a}u} f(u)\,\mathrm{d}u$$

For distribution:

$$\langle S_a T, e^{-st}\rangle = \frac{1}{a}\langle T, e^{-\frac{st}{a}}\rangle.$$

□

xvi Complex conjugate theorem

$$\mathcal{L}\{\overline{f(t)}\} = \overline{F(\bar{s})}.$$

The ROC is the same.

Proof.

$$\int_R \overline{f(t)}e^{-st}\,\mathrm{d}t = \int_R \overline{f(t)e^{-\bar{s}t}}\,\mathrm{d}t = \overline{\int_R f(t)e^{-\bar{s}t}\,\mathrm{d}t}.$$

□

xvii Convolution theorem

For functionals:

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s).$$

Proof. By Fubini's theorem,

$$\begin{aligned}\mathcal{L}\{(f * g)(t)\} &= \int_R e^{-st}\int_R f(\tau)g(t-\tau)\,\mathrm{d}\tau\,\mathrm{d}t\,, \quad u = t - \tau \\ &= \int_R e^{-s\tau} f(\tau)\,\mathrm{d}\tau \int_R e^{-su} g(u)\,\mathrm{d}u \\ &= F(s)G(s)\end{aligned}$$

□

For functional and distribution:

$$\mathcal{L}\{f * T\} = F(s)\mathcal{L}\{T\}.$$

Proof. By Fubini's theorem,

$$\begin{aligned}
\mathcal{L}\{f * T\} &= \int_R \langle T(x), f(t-x)\rangle e^{-st}\,\mathrm{d}t \\
&= \langle T(x), \int_R f(t-x)e^{-st}\,\mathrm{d}t\rangle \\
u = t - x, &\quad \mathrm{d}t = \mathrm{d}u \\
\int_R f(t-x)e^{-st}\,\mathrm{d}t &= e^{-sx}\int_R f(u)e^{-su}\,\mathrm{d}u \\
&= e^{-sx}F(s) \\
\langle T, e^{-sx}F(s)\rangle &= F(s))\mathcal{L}\{T\}
\end{aligned}$$

□

For distributions:

$$\mathcal{L}\{S * T\} = \mathcal{L}\{S\}\mathcal{L}\{T\}.$$

Proof. By Fubini's theorem,

$$\begin{aligned}
\mathcal{L}\{S * T\} &= \langle S(t), \langle T(x), e^{-s(t+x)}\rangle\rangle \\
&= \langle S(t), e^{-st}\langle T(x), e^{-sx}\rangle\rangle \\
&= \mathcal{L}\{S\}\mathcal{L}\{T\}
\end{aligned}$$

□

xviii Cross-correlation theorem for two-sided Laplace transform

For two-sided Laplace transform of functionals:

$$\mathcal{L}\left\{(f \star g)(t)\right\} = \overline{F(s)}G(s).$$

Proof. By Fubini's theorem,

$$\begin{aligned}
\mathcal{L}\left\{(f \star g)(t)\right\} &= \int_{-\infty}^{\infty} e^{-st}\int_{-\infty}^{\infty} \overline{f(-\tau)}g(t-\tau)\,\mathrm{d}\tau\,\mathrm{d}t\,, \quad u = t - \tau \\
&= \int_{-\infty}^{\infty} e^{-s\tau}\overline{f(-\tau)}\,\mathrm{d}\tau\int_{-\infty}^{\infty} e^{-su}g(u)\,\mathrm{d}u \\
&= \overline{F(s)}G(s)
\end{aligned}$$

□

TODO: For two-sided Laplace transform of functional and distribution: TODO: For two-sided Laplace transform of distributions:

xix Initial Value Theorem (IVT) for one-sided Laplace transform of functional

For functional f of expertial order and its one-sided Laplace transform $F(s)$, for $s \in \mathbb{R}$,

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s).$$

Proof. For $s \in \mathbb{R}$:

$$sF(s) = \int_0^\infty sf(t)e^{-st}\,\mathrm{d}t\,.$$

Let $u = st$, $\mathrm{d}t = \dfrac{\mathrm{d}u}{s}$.

$$sF(s) = \int_0^\infty f\left(\frac{u}{s}\right)e^{-u}\,\mathrm{d}u\,.$$

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty}\int_0^\infty f\left(\frac{u}{s}\right)e^{-u}\,\mathrm{d}u\,.$$

Define a net of functions $\langle f_s(u) = f\left(\dfrac{u}{s}\right)\rangle$ where s is in the ROC of $F(s)$.

Since $f(t)$ is in $L^1(\mathbb{R}_{\geq 0})$, $f(0^+)$ exists. For any fixed $u \in \mathbb{R}_{>0}$, $\lim\limits_{s\to\infty}\dfrac{u}{s} \to 0^+$, so $f_s(u)$ pointwise converges to $f(0^+)$, and $f_s(u)e^{-u}$ pointwise converges to $f(0^+)e^{-u}$.

For dominated convergence theorem, we require an integrable function $g(u)$ such that

$$\left|f\left(\frac{u}{s}\right)e^{-u}\right| \leq g(u)$$

a.e. for all $s > 0$.

Since $f(t)$ is in $L^1(\mathbb{R}_{\geq 0})$, there exists $C \in \mathbb{R}$ such that

$$|f(t)| \leq C \quad \forall 0 \leq t \leq T.$$

Since $f(t)$ is of exponential order, there exists $\alpha > 0$, $M > 0$, and $T > 0$ such that:

$$|f(t)| \leq Me^{\alpha t} \quad \forall t > T.$$

$$f\left(\frac{u}{s}\right)e^{-u} \leq Ce^{-u} + Me^{\alpha\frac{u}{s}}e^{-u} = Ce^{-u} + Me^{-u\left(1-\frac{\alpha}{s}\right)} \leq Ce^{-u} + Me^{-\frac{u}{2}}, \quad \forall s > 2\alpha.$$

By dominated convergence theorem, we obtain:

$$\begin{aligned}\lim_{s\to\infty}\int_0^\infty f_s(u)e^{-u}\,\mathrm{d}u &= \lim_{n\to\infty}\int_0^\infty f(0^+)e^{-u}\,\mathrm{d}u \\ &= f(0^+)\int_0^\infty e^{-u}\,\mathrm{d}u \\ &= f(0^+).\end{aligned}$$

□

xx Final Value Theorem (FVT) for one-sided Laplace transform of functional

For functional $f(t)$ with $\lim_{t\to\infty} f(t) \in \mathbb{C}$ and its one-sided Laplace transform $F(s)$ with no pole in $\{s \in \mathbb{C} \mid \Re(s) \geq 0\} \setminus \{0\}$, for $\{s \in \mathbb{C} \mid \Re(s) \geq 0\}$, with $\lim_{s\to 0} sF(s) \in \mathbb{C}$,

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s).$$

Proof. Since $\lim_{t\to\infty} f(t)$, there exists M such that $|f(t)| \leq M \in \mathbb{R}$ a.e. on (a, ∞) for some $a \geq 0$. Let

$$L = \lim_{t\to\infty} f(t).$$

For all $\varepsilon > 0$, there exists $\delta > 0$ such that $t > \delta$ implies $|f(t) - L| \leq \varepsilon$.

Since

$$s \int_0^\infty e^{-st}\,\mathrm{d}t = 1,$$

for every s,

$$sF(s) - L = s \int_0^\infty (f(t) - L)e^{-st}\,\mathrm{d}t\,.$$

Therefore,

$$\begin{aligned}
|sF(s) - L| &\leq s \int_0^\infty |f(t) - L|\, e^{-st}\,\mathrm{d}t \\
&\leq 2Ms \int_0^\delta e^{-st}\,\mathrm{d}t + \varepsilon s \int_\delta^\infty e^{-st} \\
&\leq 2M\left(1 - e^{-s\delta}\right) + \varepsilon s \int_0^\infty e^{-st} \\
&\leq 2Ms\delta + \varepsilon
\end{aligned}$$

□

xxi Cosine and sine functions

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}.$$

Proof.

$$\begin{aligned}
\mathcal{L}\{\cos(\omega t)\} &= \frac{1}{2}\left(\mathcal{L}\{e^{i\omega t}\} + \mathcal{L}\{e^{-i\omega t}\}\right) \\
&= \frac{1}{2}\left(\frac{1}{s - i\omega} + \frac{1}{s + i\omega}\right) \\
&= \frac{1}{2}\frac{2s}{s^2 + \omega^2} \\
&= \frac{s}{s^2 + \omega^2}
\end{aligned}$$

□

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\sin(\omega t)\} &= \frac{1}{2i}\left(\mathcal{L}\{e^{i\omega t}\} - \mathcal{L}\{e^{-i\omega t}\}\right)\\ &= \frac{1}{2i}\left(\frac{1}{s-i\omega} - \frac{1}{s+i\omega}\right)\\ &= \frac{1}{2i}\frac{2i\omega}{s^2+\omega^2}\\ &= \frac{\omega}{s^2+\omega^2}\end{aligned}$$

□

$$\mathcal{L}\{\cos(\omega t - \varphi)\} = \frac{\cos\varphi s + \sin\varphi\omega}{s^2+\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\cos(\omega t - \varphi)\} &= \mathcal{L}\{\cos\varphi\cos(\omega t) + \sin\varphi\sin(\omega t)\}\\ &= \frac{\cos\varphi s + \sin\varphi\omega}{s^2+\omega^2}\end{aligned}$$

□

$$\mathcal{L}\{\sin(\omega t - \varphi)\} = \frac{\cos\varphi\omega - \sin\varphi s}{s^2+\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\sin(\omega t - \varphi)\} &= \mathcal{L}\{\cos\varphi\sin(\omega t) - \sin\varphi\cos(\omega t)\}\\ &= \frac{\cos\varphi\omega - \sin\varphi s}{s^2+\omega^2}\end{aligned}$$

□

xxii Hyperbolic cosine and sine functions

$$\mathcal{L}\{\cosh(\omega t)\} = \frac{s}{s^2-\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\cosh(\omega t)\} &= \frac{1}{2}\left(\mathcal{L}\{e^{\omega t}\} + \mathcal{L}\{e^{-\omega t}\}\right)\\ &= \frac{1}{2}\left(\frac{1}{s-\omega} + \frac{1}{s+\omega}\right)\\ &= \frac{1}{2}\frac{2s}{s^2-\omega^2}\\ &= \frac{s}{s^2-\omega^2}\end{aligned}$$

□

$$\mathcal{L}\{\sinh(\omega t)\} = \frac{\omega}{s^2-\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\sinh(\omega t)\} &= \frac{1}{2}\left(\mathcal{L}\{e^{\omega t}\} - \mathcal{L}\{e^{-\omega t}\}\right)\\ &= \frac{1}{2}\left(\frac{1}{s-\omega} - \frac{1}{s+\omega}\right)\\ &= \frac{1}{2}\frac{2\omega}{s^2-\omega^2}\\ &= \frac{\omega}{s^2-\omega^2}\end{aligned}$$

□

$$\mathcal{L}\{\cosh(\omega t - \varphi)\} = \frac{\cosh\varphi s - \sinh\varphi\omega}{s^2-\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\cosh(\omega t - \varphi)\} &= \mathcal{L}\{\cosh\varphi\cosh(\omega t) - \sinh\varphi\sinh(\omega t)\}\\ &= \frac{\cosh\varphi s - \sinh\varphi\omega}{s^2-\omega^2}\end{aligned}$$

□

$$\mathcal{L}\{\sinh(\omega t - \varphi)\} = \frac{\cosh\varphi\omega - \sinh\varphi s}{s^2-\omega^2}.$$

Proof.

$$\begin{aligned}\mathcal{L}\{\sinh(\omega t - \varphi)\} &= \mathcal{L}\{\cosh\varphi\sinh(\omega t) - \sinh\varphi\cosh(\omega t)\}\\ &= \frac{\cosh\varphi\omega - \sinh\varphi s}{s^2-\omega^2}\end{aligned}$$

□

xxiii Power function

$$\mathcal{L}\{t^q\} = \frac{q!}{s^{q+1}}, \quad q \in \mathbb{N}_0.$$

$$\mathcal{L}\{t^q\} = \frac{\Gamma(q+1)}{s^{q+1}}, \quad \Re(q) > -1.$$

The ROC is $\Re(s) > 0$.

Proof. Let $u = st, \quad \mathrm{d}u = s\,\mathrm{d}t$.

$$\begin{aligned}\mathcal{L}\{t^q\} &= \frac{1}{s}\int_0^\infty e^{-u}\left(\frac{u}{s}\right)^q \mathrm{d}u\\ &= \frac{1}{s^{q+1}}\int_0^\infty u^q e^{-u}\,\mathrm{d}u\\ &= \frac{\Gamma(q+1)}{s^{q+1}}\end{aligned}$$

□

xxiv Mellin's inverse formula, Bromwich integral, or (Fourier–Mellin integral of) inverse Laplace transform

The inverse Laplace transform of a complex function $F(s)$, denoted as $\mathcal{L}^{-1}\{F(s)\}(t)$ or $f(t)$, is a complex-valued functional over the reals defined as

$$\mathcal{L}^{-1}\{F(s)\}(t) = f(t) = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} F(s)e^{st}\,\mathrm{d}s\,,$$

where γ is any real number such that it is greater than the real parts of all singularities of F and that F is bounded on the line $s = \gamma$.

For any complex-valued functional f defined on $\mathbb{R}$ such that $f(t)e^{-\gamma t} \in L^1(\mathbb{R})$, the inverse Laplace transform of the two-sided Laplace transform of $f(t)$ is equal to $f(t)$ a.e.

Proof. We want to show that:

$$f(t) = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}\int_{-\infty}^{\infty} f(u)e^{-su}\,\mathrm{d}u\,\mathrm{d}s$$

a.e.

By Fubini's theorem:

$$\frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}\int_{-\infty}^{\infty} f(u)e^{-su}\,\mathrm{d}u\,\mathrm{d}s = \lim_{T\to\infty}\int_{-\infty}^{\infty} f(u)\frac{1}{2\pi i}\int_{\gamma-iT}^{\gamma+iT} e^{s(t-u)}\,\mathrm{d}s\,\mathrm{d}u\,.$$

Let

$$K_T(t-u) = \frac{1}{2\pi i}\int_{\gamma-iT}^{\gamma+iT} e^{s(t-u)}\,\mathrm{d}s\,.$$

Let

$$s = \gamma + i\omega.$$

$$K_T(\tau) = \frac{e^{\gamma\tau}}{2\pi}\int_{-T}^{T} e^{i\omega\tau}\,\mathrm{d}\omega = e^{\gamma\tau}\frac{\sin(T\tau)}{\pi\tau}.$$

Let

$$g(u) = f(u)e^{-\gamma u}.$$

Let

$$v = t - u.$$

$$\begin{aligned}
\lim_{T\to\infty}\int_{-\infty}^{\infty} f(u)\frac{1}{2\pi i}\int_{\gamma-iT}^{\gamma+iT} e^{s(t-u)}\,\mathrm{d}s\,\mathrm{d}u &= \lim_{T\to\infty}\int_{-\infty}^{\infty} f(u)e^{\gamma(t-u)}\frac{\sin(T(t-u))}{\pi(t-u)}\,\mathrm{d}u \\
&= e^{\gamma t}\lim_{T\to\infty}\int_{-\infty}^{\infty} g(u)\frac{\sin(T(t-u))}{\pi(t-u)}\,\mathrm{d}u \\
&= e^{\gamma t}\lim_{T\to\infty}\int_{-\infty}^{\infty} g(t-v)\frac{\sin(Tv)}{\pi v}\,\mathrm{d}v \\
&= e^{\gamma t}g(t) \\
&= f(t)
\end{aligned}$$

□

For any complex-valued functional f defined on $\mathbb{R}$ such that $f(t) = 0$ for all $\Re(t) < 0$ and $f(t)e^{-\gamma t} \in L^1(\mathbb{R}_{\geq 0})$, the inverse Laplace transform of the one-sided Laplace transform of $f(t)$ is equal to $f(t)$ a.e.

xxv Inverse Laplace transform of constant

For any $c \in \mathbb{C}$,

$$\mathcal{L}^{-1}\{c\} = 0$$

Proof. For $c = 0$, it is obvious.

For $c \neq 0$,

$$\begin{aligned}
\lim_{T\to\infty} \int_{-iT}^{iT} e^{st}\,\mathrm{d}s &= \lim_{T\to\infty} \left(\frac{1}{s}e^{st}\right)\Bigg|_{-iT}^{iT} \\
&= \lim_{T\to\infty} \frac{e^{iT} + e^{-iT}}{iT} \\
&= \lim_{T\to\infty} \frac{2\cos(T)}{iT} \\
&= 0
\end{aligned}$$

□

xxvi Cover-up method for partial fraction decomposition

Suppose $P(s)$ is a polynomial of s and is not divisible by any of $(s - a_i)$, $a_i \in \mathbb{C}, b_i \in \mathbb{N}$ and $i \neq j \implies a_i \neq a_j$, and

$$\frac{P(s)}{\prod_{i=1}^{n}(s-a_i)^{b_i}} = \sum_{i=1}^{n}\sum_{j=1}^{b_i} \frac{A_{i,j}}{(s-a_i)^j}.$$

Then

$$A_{k,b_k-m} = \frac{1}{m!}\left\{\frac{\mathrm{d}^m}{\mathrm{d}s^m}\left(\frac{P(s)}{\prod_{\substack{i=1\\ i\neq k}}^{n}(s-a_i)^{b_i}}\right)\right\}\Bigg|_{s=a_k}.$$

xxvii Transform of a periodic function

Suppose $f(t)$ is periodic with period $T > 0$ and of exponential order.

For one-sided Laplace transform:

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\,\mathrm{d}t$$

Proof.

$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t)\,\mathrm{d}t + \int_T^\infty e^{-st} f(t)\,\mathrm{d}t$$

Let $t = u + T$.

$$\int_T^\infty e^{-st} f(t)\,\mathrm{d}t = \int_0^\infty e^{-s(u+T)} f(u+T)\,\mathrm{d}u = e^{-sT}\mathcal{L}\{f(t)\}$$

□

For two-sided Laplace transform: The two-sided Laplace transform of a periodic function of period T such that $\int_0^T e^{-su} f(u)\,\mathrm{d}u \neq 0$ must diverge.

Proof.

$$\mathcal{L}\{f(t)\} = \sum_{k=-\infty}^{\infty} \int_{kT}^{(k+1)T} e^{-st} f(t)\,\mathrm{d}t$$

Let $u = kT$.

$$\sum_{k=-\infty}^{\infty} \int_{kT}^{(k+1)T} e^{-st} f(t)\,\mathrm{d}t = \int_0^T e^{-su} f(u)\,\mathrm{d}u \sum_{k=-\infty}^{\infty} e^{-skT}$$

$\sum_{k=-\infty}^{\infty} e^{-skT}$ diverges for all s. □

III Fourier Transform (FT) (傅立葉變換)

TODO TODO: mb as Laplace transform special case?

i Fourier Transform (FT)

Fourier transform is an integral transform that converts a function of a real variable (usually t, in the time domain) to a function of another real variable (usually ω, in the frequency or Fourier domain). The functions are often denoted in lowercase for the time-domain representation and uppercase for the frequency-domain.

The Fourier transform, denoted as $\mathcal{F}\{f(t)\}(\omega)$ or $F(\omega)$, is defined by the improper integral

$$\mathcal{F}\{f(t)\}(\omega) = F(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t)\,\mathrm{d}t\,.$$

ii Inverse Fourier Transform (反傅立葉變換)

TODO

IV Dirichlet kernel

TODO